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# 337635 (37)

BE (6th Semester)

Examination, Nov.-Dec., 2013

Branch : Mechanical

## POWER PLANT ENGINEERING

*Time Allowed : Three Hours*

*Maximum Marks : 80*

*Minimum Pass Marks : 28*

lote : All questions carry equal marks. Solve any two parts from each unit.

### UNIT - I

. 1. (a) What are the various types of power stations ? Explain with diagram construction and working of modern steam power plant. 8

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(b) Describe the elements of electric pow systems in brief.

(c) Discuss the importance of central pow stations.

### UNIT - II

l. 3.

Q. 2. (a) Name the various methods of ash handlir Why it is essential to quench the ash befo handling. Describe any two methods of a handling.

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(b) What do you mean by heat balance sheet ?

Write the heat balance sheet for a boiler. 8

(c) Write short notes on :

(i) Air pre-heater 4

(ii) Feed water treatment 4

UNIT - III

Q. 3. (a) With the help of a simple diagram, explain the essential features of a hydro, electric power plant. 8

(b) Compare hydro-electric power stations and thermal power stations. 8



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(c) Discuss the advantages and disadvant of a diesel power plant. State the applic of diesel power plant ?

UNIT - IV

Q. 4. (a) Draw a neat diagram of a reactor and e the functions of its various components

(b) What is a chain reaction ? How it is contr Differentiate between uncontrolled controlled chain reactions.

(c) Discuss the safety aspects of a nuclear

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UNIT - V

Q. 5. (a) What is meant by load curve ? Explain its importance in power generation. 8

(b) Define 'average load', 'peak load', and 'base load'. A hydro power plant is to be used as peak load plant at an annual load factor of 30%. The electrical energy obtained during the year is $750 \times 10^5$ kWh. Determine the maximum demand. If the plant capacity factor

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is 24%, find reserve capacity of the plant. Take 8760 working hours in a year. 8

(c) Write short notes on : 4×2

(i) Depreciation and replacement policy

(ii) Cost of plant and production.



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